

Toward a Geometric Perspective on Trustworthy AI: Diagnostic Signatures for High-Stakes NLU

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The deployment of Natural Language Understanding (NLU) systems in high-risk sectors such as clinical triage, emergency response, and financial eligibility, is now governed by the transparency and accountability mandates of the European AI Act. Despite these legal obligations, a significant "Accountability Gap" persists: regulators demand documentation of a system's "general logic", while the deep learning models powering these systems remain structurally opaque. While existing Explainable AI (XAI) methods provide post-hoc, model-agnostic approximations, they often describe where a model attended rather than how it computed its decision, a distinction that is increasingly consequential in high-stakes settings. This paper proposes that the Geometric Framework for Intent Detection offers a complementary analytical lens for addressing these challenges. By interpreting semantic resolution as a dynamical trajectory through a hidden state manifold converging toward stable attractor basins, this framework makes visible certain structural properties of the model's computation that are currently invisible to behavioral audits. We argue that this geometric perspective opens a set of promising research directions for each of the five pillars of Trustworthy AI: Transparency, Technical Robustness, Fairness, Human Oversight, and Societal Well-being. Specifically, it suggests candidate diagnostic signatures for failures such as algorithmic bias, adversarial vulnerability, out-of-scope instability, and multilingual degradation. We present this framework as a conceptual toolbox and a research roadmap intended to complement existing auditing practices and contribute toward the goal of verifiable, mechanistically grounded NLU.

Keywords: Trustworthy AI, NLU, Intent Detection, RNNs

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1 Introduction

Modern AI-driven interaction is built upon dialogue systems, where NLU serves as the critical bridge between human intent and automated action. As these systems transition into high-risk systemic domains, such as clinical triage [12], financial eligibility assessment [6], and emergency response [7], the requirements for their deployment have undergone a fundamental shift. Under the European AI Act [8], high-risk AI systems are no longer evaluated

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solely on predictive success. They are now subject to rigorous mandates for transparency, robustness, and accountability. This disconnect between the legal demand for explainability and the technical opacity of deep learning [22] is what this paper terms the *Accountability Gap*. This gap is not merely a documentation inconvenience but a structural barrier to the production of the technical evidence required for high-risk certification.

The field of Explainable AI (XAI) has sought to mitigate this opacity through post-hoc, model-agnostic methods like SHAP [20] and LIME [25]. While these tools provide valuable attributive explanations by identifying influential input features, they remain approximations of behavior rather than descriptions of mechanism [28]. They describe what the model attended to, not how it arrived at its final answer. For researchers and policy-makers working in high-stakes environments, this raises a legitimate concern: whether attributive approximations provide the mechanistic grounding that verifiable AI governance ultimately demands [11].

This paper proposes that the Geometric Framework for Intent Detection [30] represents a valuable, and largely underexplored, complementary direction for addressing this challenge. By treating Recurrent Neural Networks (RNNs) as non-linear dynamical systems, the framework offers a window into the internal geometry through which semantic resolution occurs. In a well-trained model, trajectories through the high-dimensional state space converge toward stable, localized regions representing specific intents [29]. This geometric organization is a measurable structural property [31] that may offer diagnostic information that behavioral testing alone cannot provide.

We present this work as a conceptual bridge and a research roadmap, arguing that the geometric perspective opens a productive set of research questions about trustworthy AI. While significant challenges remain, including the transition to Transformer architectures [36], the framework provides a new diagnostic language for identifying "geometric pathologies" in NLU systems that manage essential human services.

2 Our Contributions and Paper Organization

The central argument of this paper is that the Geometric Framework for Intent Detection [29–31] offers a complementary analytical lens that may help bridge the Accountability Gap in high-risk NLU. Rather than proposing a final or singular solution, the work aims to show that viewing the pillars of Trustworthy AI through a geometric lens, interpreting internal states as manifolds and trajectories, reveals a concrete set of research questions currently inaccessible to behavioral audits.

This paper presents a conceptual toolbox that identifies "geometric pathologies" in systems managing essential services, investigating the potential for mechanistic evidence to eventually inform the technical documentation required for high-risk AI certification. The specific contributions are as follows:

- **A Geometric Reinterpretation of Regulatory Pillars.** The paper proposes a reinterpretation of each of the five pillars of the EU AI Act's Trustworthy AI framework as a set of structural questions about the model's hidden state manifold. This approach suggests that geometric analysis can provide a mechanistic perspective to complement existing behavioral and post-hoc XAI methods.
- **Formalization of a Candidate Diagnostic Vocabulary.** Building on empirical findings from prior work, this paper introduces a vocabulary of geometric "pathologies" including *Geometric Collapse*, *Readout Misalignment*, *Geometric Leakage*, *Topological Deviation*, as candidate diagnostic categories for failure modes currently difficult to detect through output-level auditing alone.

- **Preliminary Structural Evidence Benchmarks.** This paper identifies a set of geometric properties, such as the Silhouette Score per intent class, centroid-readout alignment matrices, Functional Dimensionality, and Distillation Efficiency ratio, as candidate metrics for structural audit documentation in high-stakes settings.
- **A Research Roadmap for Mechanistic Safety.** This work consolidates identified open problems into a structured roadmap. This roadmap outlines methodological and empirical work needed to transition geometric toolbox from an analytical state toward potential applicability in real-world high-risk certification contexts.

The remainder of the paper is organized as follows: Section 3 contextualizes Intent Detection within the NLU pipeline; Section 4 outlines the regulatory landscape and reframes the pillars of Trustworthy AI as mechanistic research questions; Section 5 establishes the technical foundations of the Geometric Framework; Section 6 through 9 explore the potential contributions of geometric analysis to each pillar; Section 10 presents the integrated research roadmap; and Section 11 concludes the work.

3 Intent Detection within the NLU Pipeline

To situate the discussion of Trustworthy AI challenges, it is necessary to understand the functional role of Intent Detection within the broader Spoken Language Understanding (SLU) architecture [35]. As illustrated in Figure 1, NLU block functions as a multi-stage semantic gateway, transforming raw linguistic inputs into actionable commands. Following speech recognition (ASR), the pipeline typically identifies the domain (e.g. a clinical query), detects the specific intent (e.g. acute symptom report), and extracts entities via slot filling (e.g. which symptom) [23]. Of these components, Intent Detection carries the highest systemic risk, an error at this node propagates through the entire pipeline, rendering perfectly extracted entities functionally irrelevant if the overarching objective is misidentified. In a clinical triage system, for instance, misclassifying a report of "acute cardiac event" as "general scheduling inquiry" constitutes a "silent failure" [21]. In such cases, the system may act confidently and fluently on a catastrophically wrong premise, with no external signal indicating that a representational error has occurred.

Under Article 6(2) and Annex III of the EU AI Act, systems managing access to essential services, including emergency healthcare triage and financial eligibility assessment, are classified as high-risk. This classification triggers a documentation burden that creates a fundamental tension: the most capable architectures, such as RNNs [24] and Transformers [1], are also those whose internal decision logic is least accessible to inspection [3]. Standard evaluation methods (e.g. F1-scores, accuracy) confirm statistical success but provide no description of the process through which individual decisions are reached.

While post-hoc XAI methods partially address this opacity by identifying influential input features [20, 25], these attributive explanations describe the model's behavior in terms of its inputs and outputs rather than its internal logic. It remains an open question whether this level of description satisfies the documentation requirements of Article 11 [4]. Behavioral testing can demonstrate that a system passed a benchmark, but it provides limited visibility into structural failure modes, such as a model that achieves high accuracy while routing minority group utterances through internally less stable representations, a fragility that would not be apparent from examining any standard performance report. The Geometric Framework is proposed here as a complementary analytical layer that examines the model's internal organization to illuminate these otherwise invisible risks.

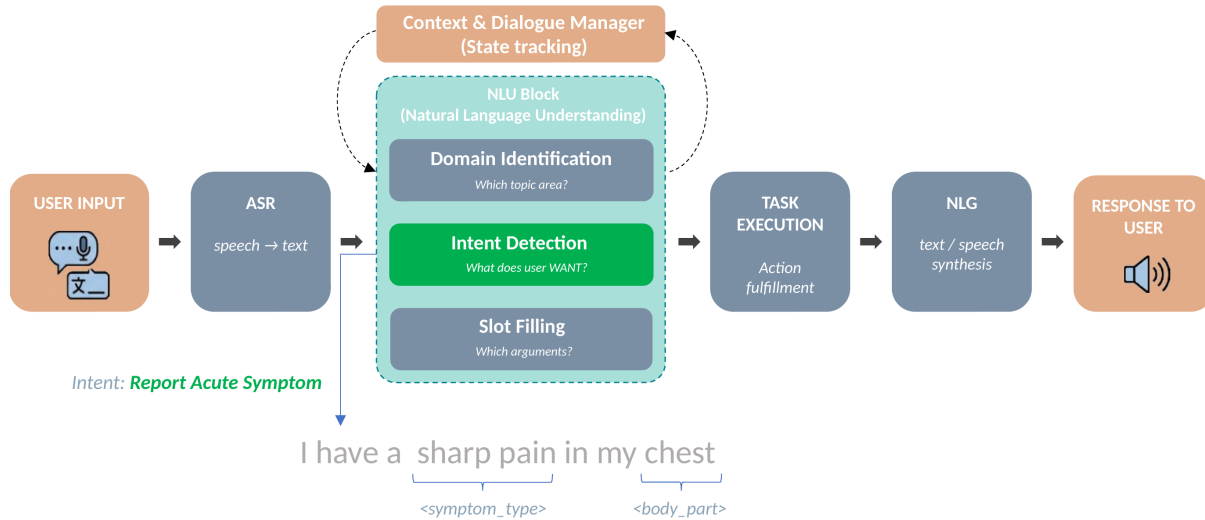


Fig. 1. **The SLU Pipeline as a Semantic Gateway.** High-risk clinical triage example shows how Intent Detection acts as the primary-decision node. A failure to correctly identify the *Report_Acute_Symptom* intent would propagate through the system.

4 Regulatory Landscape: Defining the Pillars of Trustworthy AI

The governance of Artificial Intelligence in Europe has transitioned from the voluntary ethical guidelines of the 2019 High-Level Expert Group [15] toward a mandatory, risk-based regulatory framework centered on the protection of fundamental rights and safety. For the NLU systems discussed in Section 3, the EU AI Act [8] activates a specific set of obligations organized around five foundational pillars of Trustworthy AI. Each pillar targets a different dimension of system quality and translates abstract principles into concrete technical evidence requirements that providers must satisfy before deployment.

- **Pillar I: Transparency and Documentation of Algorithmic Logic.** Article 11 mandates technical documentation describing the "general logic of the AI system and of the algorithms". For deep learning NLU, where decision logic is distributed across high-dimensional parameter spaces rather than explicit rules, providing such insight remains a significant challenge [9]. Current post-hoc methods characterize behavior in terms of inputs and outputs, offering limited visibility into how a model resolves specific user intents [28]. The Geometric Framework proposes that trajectory-level analysis of the hidden state space may offer a complementary form of process documentation (Section 6).
- **Pillar II: Technical Robustness and Safety.** Article 15 requires that high-risk systems maintain resilience not only under normal operating conditions but under foreseeable adversarial or distributional stress. A well-documented failure in this regard is shortcut learning [10], where models achieve high benchmark accuracy by exploiting brittle surface (linguistic) correlations rather than robust semantic understanding [21], whose structural fragility only becomes apparent under distribution shift or adversarial perturbation [26]. The Geometric Framework suggests that certain properties of the model's internal representational organization may carry structural information about these vulnerabilities (Section 7).

- **Pillar III: Fairness and Non-Discrimination.** Article 10 requires that training data and model behavior be free from properties that produce discriminatory outcomes for users based on characteristics such as dialect, accent, or linguistic variety [5]. Current fairness auditing is primarily retrospective, measuring performance disparities after they have emerged in aggregate statistics [34]. This approach is valuable but provides limited guidance on the internal conditions that produce discrimination or the interventions that could prevent it. The Geometric Framework proposes a complementary, structurally oriented perspective, formalized through three candidate pathologies (Section 8).
- **Pillar IV: Human Oversight and Actionable Diagnostic.** Article 14 stipulates that high-risk systems be designed to enable effective oversight by natural persons, supporting timely and targeted intervention. In practice, human supervisors are typically equipped with a classification label and confidence score [2], information sufficient to flag a suspicious output but insufficient to distinguish between fundamentally different categories or failure, each requiring a different corrective action. The Geometric Framework proposes a mechanistic decomposition of failure that could support more diagnostic and proportionate oversight (Section 6).
- **Pillar V: Societal and Environmental Well-being.** Article 69 extend regulatory concern toward two distinct systemic requirements. The first is linguistic inclusion: safety and performance properties must remain stable across the Union’s diverse languages. The second is environmental sustainability, AI infrastructure must minimize its energy footprint, The Geometric Framework addresses these by identifying low-dimensional subspaces where computations are concentrated. Preliminary results suggest a cross-lingual stability of these subspaces. Furthermore, the identification of task-relevant subspaces could allow to prune models to their functional minimum without degrading safety profiles required for high-risk systems. (Section 9).

5 Technical Framework: A Geometric View of Semantic Resolution

This section establishes a formal vocabulary for describing the system’s internal behavior in a way that is mathematically precise and empirically measurable. The four principles defined here: trajectory logic, semantic cluster formation, readout alignment, and low-dimensional structure form the analytical foundation from which all diagnostic categories, candidate metrics, and research directions in Sections 6 through 9 are derived. The material summarizes foundational findings from prior work [29–31].

5.1 Ambient Space and Trajectory Logic

An RNN processes an input sequence of tokens $\mathbf{x}_1, \dots, \mathbf{x}_T$ by maintaining an internal memory vector, the hidden state $\mathbf{h}_t \in \mathbb{R}^D$ updated at each timestep via a non-linear transition function $\mathbf{h}_t = \mathbf{F}(\mathbf{h}_{t-1}, \mathbf{x}_t)$ (Figure 2a). The parameter D is the number of hidden units, an engineering parameter set at design time that defines the model’s *ambient space* (a.k.a. ambient capacity), the total dimensionality of the space in which model’s internal states live. The framework’s central observation is that full sequence of hidden states $(\mathbf{h}_1, \dots, \mathbf{h}_T)$ traces a *trajectory* through this ambient space as the model processes each input token (Figure 2b). Each token displaces the hidden state in some direction and the effect of the entire utterance is encoded in the final state $\mathbf{h}_T$, from which the classification is derived. In this view, the trajectory is the most direct computational record of how the model processed the input, not a summary of its behavior, but the behavior itself, unfolding step by step through the representational space.

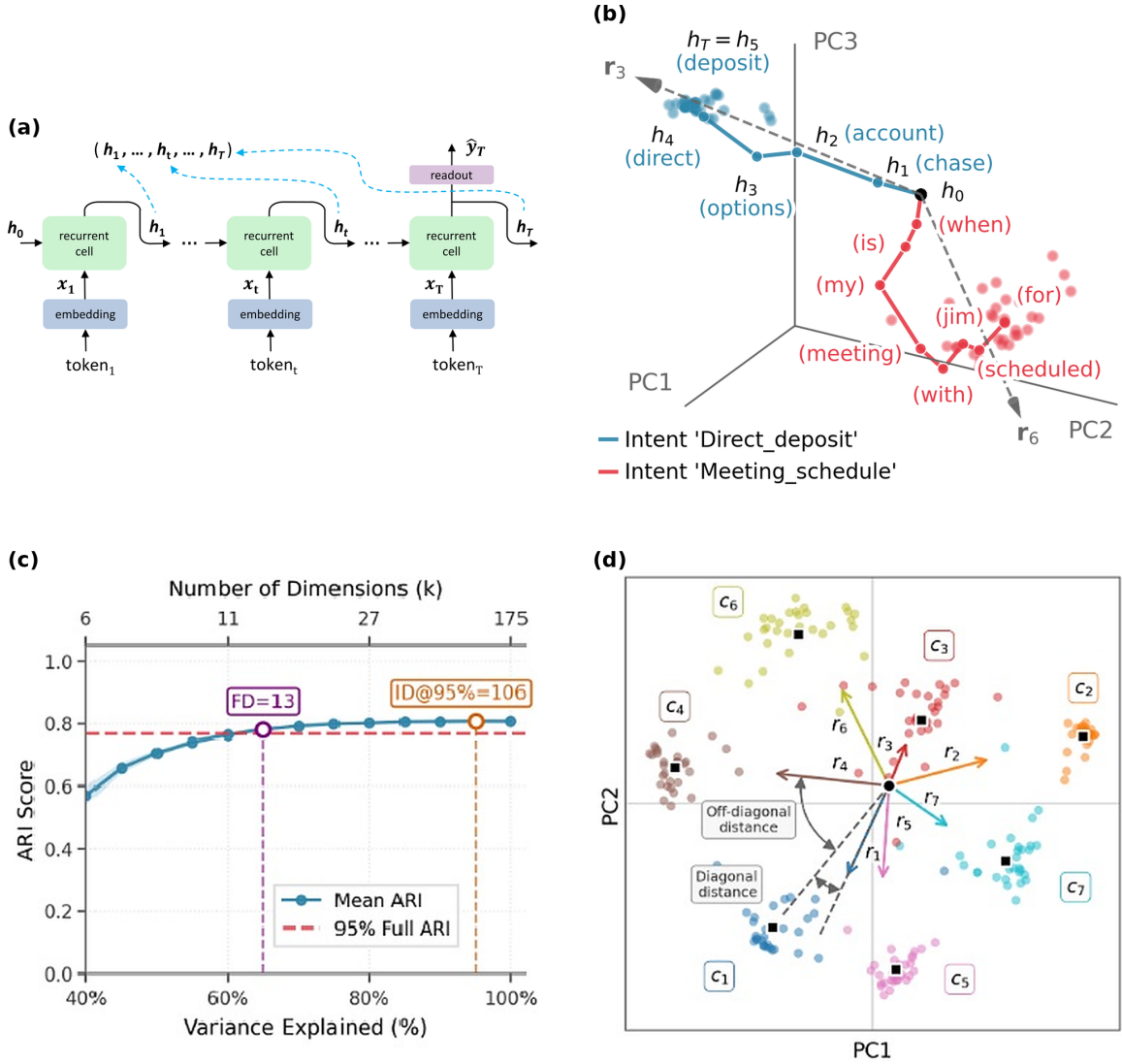


Fig. 2. **Mechanistic Principles of the Geometric Framework.** (a) Architectural trace of the recurrent update process $\mathbf{h}_t = \mathbf{F}(\mathbf{h}_{t-1}, \mathbf{x}_t)$. (b) 3D projection of hidden state trajectories converging toward stable attractors. (c) Comparison between intrinsic dimensionality (*ID*) and hyper-efficient Functional Dimensionality (*FD*) subspace. (d) Semantic cluster organization and angular alignment with readout vectors $\mathbf{r}_j$, supporting the mechanistic diagnosis of representation and decision failures.

5.2 Semantic Cluster Formation

Empirical analysis of RNNs trained for intent detection reveals that the final hidden states $\mathbf{h}_T$ are not distributed arbitrarily across $\mathbb{R}^D$. Instead, they organize into coherent, localized regions called *semantic clusters* corresponding

to specific intent classes. In dynamical systems terms, these regions behave as attractors, stable configurations toward which trajectories belonging to the same intent class converge, regardless of surface-level variation in the input. The center of each attractor is the *semantic centroid* ($\mathbf{c}_j$) for intent class j (Figure 2b).

The structural cohesion of each attractor is measurable via the Silhouette Score [27] computed per intent class. A value near +1 indicates a tight, well-separated cluster, a geometrically stable representation of the intent. Conversely, a low or negative score indicates what we term *Geometric Collapse*, a structural condition in which the model fails to form a distinct representation for that intent class. In controlled studies [30] on the ATIS dataset [13], this collapse was observed systematically in low-frequency intent classes, whose hidden states dissolved into an undifferentiated region of the state space rather than converging to a distinct cluster, a failure mode that could not be identified by examining the model’s outputs alone.

5.3 Readout Alignment and the Inference Process

Classification is produced when the final hidden state $\mathbf{h}_T$ is projected through a linear readout layer $\mathbf{W} \in \mathbb{R}^{N \times D}$, where N is the number of intent classes. The rows of $\mathbf{W}$, denoted, $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$, are the readout vectors that define the decision boundaries. The predicted class is $i^* = \operatorname{argmax}_i(\mathbf{r}_i, \mathbf{h}_T)$.

In well-trained models, a consistent geometric regularity emerges, the semantic centroid $\mathbf{c}_j$ tends to be angularly aligned with its own readout vector $\mathbf{r}_j$ and approximately orthogonal to all others (Figure 2d). Quantified via the $N \times N$ cosine similarity matrix between centroids and readout vectors, this structure produces a strongly diagonal alignment matrix, high similarity between each $\mathbf{c}_j$ and its correct $\mathbf{r}_j$, with off-diagonal similarity centered near zero. This pattern has been observed consistently across HWU64 [19], BANKING77 [6], and CLINC150 [18] at scales up to 150 intent classes [31]. This readout projection is the only point at which the model’s internal geometry becomes an observable classification. Everything prior, how the model reached $\mathbf{h}_T$, how well-structured that state is, whether it reliably encodes the user’s intent is invisible to standard behavioral audit.

A central claim of Geometric framework is that analyzing internal organization directly, rather than approximating it from inputs and outputs, may offer diagnostic information that behavioral methods cannot easily access. This observation motivates a decomposition of classification failure into two structurally distinct candidate categories:

- **Geometric Collapse** (Representation Error): The cluster $\mathbf{c}_j$ is poorly formed or diffuse, the model has not learned an adequate internal representation for the intent class. This points to a data-level problem, insufficient or structurally inadequate training coverage.
- **Readout Misalignment** (Decision Error): The cluster $\mathbf{c}_j$ is well-formed, but the readout vector $\mathbf{r}_j$ is incorrectly oriented, placing the decision boundary in the wrong region despite an adequate internal representation. This points to a model-level problem, potentially addressable without full retraining.

The ATIS analysis in [30] identified four empirically distinct combinations of these two conditions, including a case where strong readout alignment compensates a collapsed cluster, producing correct classification despite a representational failure that would be entirely invisible to any output-level audit. While informative as a research hypothesis, this proposal requires careful empirical validation before can be considered a reliable diagnostic tool.

5.4 Linguistic Packaging vs. Semantic Resolution

A consistent finding across prior work is that while the ambient space D may be large, the intent-relevant computation appears to occur in a much smaller subspace. This is captured by two complementary metrics:

- **Intrinsic Dimensionality (ID):** The minimum number of principal components required to explain 95% of the hidden state variance, ID is interpreted as "Linguistic Packaging", the computational volume the model allocates to surface-level structural and morphological variation in the input.
- **Functional Dimensionality (FD):** A task-aware metric identifying the minimum number of principal components required to preserve semantic cluster quality. FD represents Semantic Resolution, the compact subspace where intent discrimination actually takes place. FD is identified by projecting hidden states onto successive [17] subspaces of increasing dimension and locating the elbow at which cluster quality saturates.

As shown in Figure 2c, a model with an ambient capacity $D = 175$ and an ID of 106 may resolve 150 distinct intents in a subspace of only $FD = 13$ [31]. The ratio $\Omega = FD/ID$ which we term the *Distillation Efficiency*, quantifies how concentrated the model's task relevant computation is relative to its total representational volume.

A related observation is a Sub-Linear Scaling Pattern, as the number of intent classes grows, FD grows significantly more slowly than the number of classes, suggesting that models learn increasingly efficient representational strategies rather than dedicating independent dimensions to each new intent. We propose this trend as a candidate structural indicator of generalization quality, a potential complement to behavioral benchmarks for monitoring model health. Whether deviations from this trend reliably predict operational degradation, and at what scale such deviations become significant, requires empirical study across diverse real-world deployment conditions.

6 Mechanistic Traceability: Bridging Transparency and Human Oversight

The transparency mandate of Article 11 and the oversight requirement of Article 14 share a common technical prerequisite, the ability to access a model's decision process in a form that is not merely observational, but diagnostic. While Article 11 requires technical documentation of the "general logic" of the algorithm. Article 14 demands that human supervisors be equipped to intervene in a targeted a proportionate manner. Current NLU auditing practice addresses these goals by supplemental post-hoc XAI methods, which approximate logic by identifying influential input features [20, 25]. However, these methods describe the relationship between inputs and outputs while ignoring the internal computation that connects [28].

Geometric Framework proposes a complementary layer of process documentation termed *Mechanistic Traceability*. In this paradigm, the sequence of hidden states $(\mathbf{h}_1, \dots, \mathbf{h}_T)$ produced during inference serves as a literal record of model's behavior as it unfolds step-by-step through the representational space. Rather than approximating a model's reasoning from its external behavior, Mechanistic traceability treats hidden state trajectory as a verifiable structural fact. If a classification results from a trajectory converging stably into a pre-defined semantic attractor, this convergence constitutes a structural, verifiable fact about that specific computation. For any given utterance, such as a clinical report of an acute cardiac event, an auditor can extract the trajectory, verify that it converged to the intended attractor basin, and measure its distance from basin boundary as a structural safety margin.

The primary utility of this geometric perspective for human oversight lies in its ability to decompose classification failures into two mechanistically distinct categories (as formalized in Section 5), and visualized through the structural pathologies in Figure 4b. By identifying the specific geometric origin of an error, human supervisors

can move beyond binary error flags to a targeted diagnostic approach, where each failure mode necessitates a fundamentally different corrective action:

- **Geometric Collapse (Representation Error):** As defined by the cluster-integrity metrics in Section 5, the attractor basin for a given intent class is poorly formed or diffuse. This indicates a data-level deficiency where the training set provides insufficient coverage or quality to form a stable semantic representation.
- **Readout Misalignment (Decision Error):** In this scenario, the semantic cluster is geometrically stable, but the corresponding readout vector $\mathbf{r}_j$ is incorrectly oriented relative to the cluster centroid $\mathbf{c}_j$. This represents a model-level miscalibration where the decision boundary is misplaced despite an adequate internal representation (see Figure 3b).

This mechanistic distinction provides a path toward a surgical model correction. In traditional deep learning, correcting an error typically necessitates full-model retraining, which is computationally expensive and risks degrading performance on unrelated intent classes. Under a geometric regime, a verified case of Readout Misalignment could, in principle, be corrected by adjusting the specific readout vector $\mathbf{r}_j$ alone. This would allow human supervisors to fix decision boundaries for high-risk intents without the opacity and collateral risks of retraining, fulfilling the spirit of meaningful oversight mandated by Article 14. Whether such targeted interventions are stable and safe remains an open question, and addressing it constitutes a concrete priority in the research roadmap.

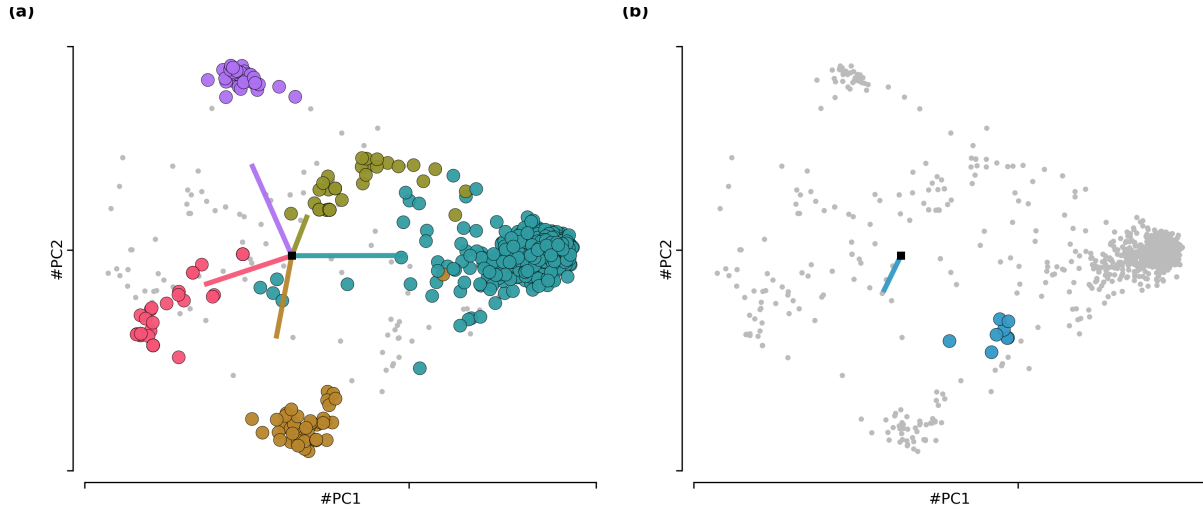


Fig. 3. **Geometric Resolution in the ATIS dataset.** (a) Optimal Organization: tight semantic clusters and precise readout alignment. (b) Structural Pathology: diffuse representations and readout misalignment.

7 Structural Integrity: Geometry of Robustness and Security

Article 15 of the EU AI Act mandates that high-risk systems maintain appropriate levels of technical robustness and resilience against foreseeable errors, including adversarial perturbations and distributional shifts. Traditionally, these challenges are addressed at the output layer. Adversarial robustness is evaluated via accuracy degradation under perturbation [16], while Out-of-Scope (OOS) detection relies on confidence thresholding of the softmax

output [14]. Both approaches share a fundamental structural limitation, they remain blind to the internal structural conditions that determine whether a model’s decision is stable or fragile. The Geometric Framework proposes a complementary perspective of these challenges, grounded in the relationship between Intrinsic Dimensionality (ID) and Functional Dimensionality (FD).

Adversarial Vulnerability and Geometric Leakage. As established in Section 5, high-performance models allocate significant representational volume to Linguistic Packaging (ID), while the actual task-solving logic is compressed into a hyper-efficient, task-aware subspace (FD). We propose the term "Geometric Leakage" to describe structural condition where excess representational volume ($ID - FD$) constitutes a potential adversarial opportunity.

The central hypothesis is that an attacker does not need to corrupt the task-relevant FD subspace to induce a "silent failure". Instead, it may be sufficient to apply a perturbation that displaces the hidden state through the "leakage" dimensions until trajectory crosses an attractor boundary. Under this mechanistic interpretation, robustness is not merely a performance statistic, but a measure of how effectively a model concentrates its logic into the minimal FD subspace. We introduce Distillation Efficiency ($\Omega = FD/ID$) as a candidate structural metric for robustness. A higher Ω implies a leaner representational strategy with fewer non-functional dimensions available for adversarial exploitation.

Out-of-Scope detection and Topological Deviation. A closely related safety requirement is the ability to recognize inputs that fall outside the system’s operational design domain [18]. For high-risk NLU, a confident misclassification is arguably more consequential than a system abstention. Current confidence-based detection methods have a structural limitation, a model can produce high-confidence softmax predictions for structurally anomalous inputs if the readout layer happens to be aligned with the out-of-distribution final hidden state.

The Geometric Framework proposes a complementary signal we term *Topological Deviation*, which monitors geometric character of the inference trajectory before final readout layer is reached. An OOS utterance, having no corresponding attractor basin in the trained manifold, is expected to produce two distinct mechanistic signatures:

- **Convergence Failure:** The trajectory fails to stabilize toward any established cluster centroid, terminating in an undifferentiated region of the state space.
- **Morphological Deviation:** Even if the trajectory terminates near a known attractor, its path (curvature and velocity profile) deviates significantly from the distribution of training trajectories associated with that class.

While these interpretations provide a grounded mechanistic lens for robustness, they define an active research frontier rather than a finished auditing methodologies.

8 Mechanistic Fairness: Patterns of Sociodemographic Bias

The non-discrimination mandate of Article 10 presents a compliance challenge that current fairness auditing addresses primarily through retrospective parity metrics, identifying performance disparities across demographic groups after they have emerged in aggregate statistics [34]. The geometric framework proposes a complementary, structurally oriented lens, bias is hypothesized as an internal representational instability in the hidden state geometry of minority group utterances, that could in principle be identified before producing discriminatory outputs.

The empirical motivation derives from class imbalance analysis in [30], showing that underrepresented classes exhibit degraded attractor structures, specifically lower Silhouette Scores and weaker readout alignment than

well-represented classes. We formalize this structural bias through three candidate pathologies of increasing severity [32], illustrated in Figure 4:

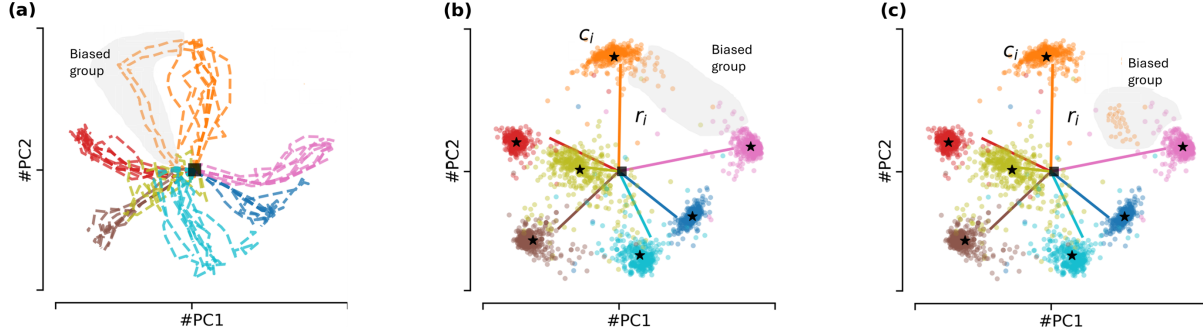


Fig. 4. **Mechanistic Pathologies of Sociodemographic Bias.** (a) Trajectory Dispersion: Minority trajectories are displaced toward basin boundaries, reducing structural safety margins. (b) Representational Contraction: Biased utterances exhibit lower Silhouette Scores. (c) Manifold Displacement: Minority utterances are systematically displaced toward incorrect attractor basins.

- **Pathology I: Trajectory Dispersion (Representational Instability).** As seen in Figure 4a), utterances from the biased group may converge toward the correct attractor basin but exhibit high trajectory dispersion. Even if the final hidden states $\mathbf{h}_T$ eventually land within the basis, the paths taken are inconsistent, high-variance, and structurally unstable compared to the majority group. These minority group utterances occupy a region where the margin from the decision boundary is consistently smaller than for majority groups. Minimal perturbations are thus sufficient to produce misclassification without affecting the majority group. This pathology can be investigated by computing per-demographic group the trajectory variance (calculated as the mean distance of each state $\mathbf{h}_t$ in a trajectory from the average path) for that intent.
- **Pathology II: Representational Contraction (Basin Weakness).** Depicted in Figure 4b), biased group utterances converge to a region with significantly lower cluster compactness. The attractor’s effective pull is weak, resulting in a diffuse representation that lacks the structural robustness of the majority classes. This fragility makes these classifications susceptible to misclassification under noise or distribution shift. As a diagnostic signal we may consider significant differences in the within-class Silhouette Score distributions when stratified by demographic group.
- **Pathology III: Manifold Displacement (Systematic Misclassification).** In this severe case (Figure 4c), the model fails to map the minority variant to its intended attractor basin. Instead, the trajectories are systematically displaced into the attractor basin of an incorrect intent. This represents a loss of representational agency, where the system consistently misinterprets the user’s intent by aligning it with the wrong readout direction. As a diagnostic signal we could consider elevated cross-class membership errors (i.e. percentage of minority-group utterances assigned to an intent cluster $j \neq \text{true_intent}$) or increased off-diagonal density in the cluster-readout alignment matrix.

These three pathologies differ not only in severity but in the type of intervention they imply. Identifying which pathology is present before deployment would allow a more targeted and proportionate corrective response. If

validated empirically, these structural signals would motivate a research direction in Geometric Regularization for Fairness, incorporating geometric fairness constraints: penalizing dispersion variance or differences in per-demographic group Silhouette Scores directly into the training loss.

9 Semantic Universality: Linguistic Inclusion & Sustainability

The societal and Environmental Well-being pillar of the EU AI Act encompasses two technically distinct yet structurally related requirements for high-risk NLU systems. The first is linguistic inclusion, which demands stable performance across the Union’s diverse languages. The second is environmental sustainability, mandating the minimization of the energy footprint of AI infrastructure. The Geometric Framework addresses both challenges by shifting the analytical focus from surface-level linguistic features to the stable core of semantic resolution.

For linguistic inclusion, the framework proposes a hypothesis termed *Resolution Stability*. This principle posits that the dimensionality of semantic resolution is governed by the task’s semantic complexity (cardinality) rather than the morphological or syntactic structure of the input language. Consequently, while Linguistic Packaging (*ID*), which captures surface-level morphological variation, varies across typologically diverse languages, the *FD* subspace remains stable [33]. If validated, this implies that the structural safety profile of a model’s semantic resolution is a language-independent property.

Regarding environmental sustainability, the consistent gap between *ID* and *FD* reveals that a substantial fraction of a model’s representational capacity is systematically devoted to Linguistic Packaging rather than to the core classification task. This structural inefficiency motivates a novel approach to Green AI: geometry-guided model compression. We hypothesize that compressing recurrent layers via Singular Value Decomposition (SVD) to a rank equal to the measured *FD* can preserve classification quality while substantially reducing energy and memory requirements. Because *FD* is a task-grounded, empirically measured ceiling rather than an arbitrary engineering parameter, this compression strategy is principled rather than heuristics, enabling high-risk NLU systems to be deployed on low-power edge without compromising the structural safety profile required by safety standards.

The Resolution Stability hypothesis and *FD*-guided compression represent significant opportunities for future research. Extending these analyses across broader language typologies could provide a formal basis for Multilingual Certification, where structural stabilities, rather than behavioral benchmarks alone, serve as evidence for the safety of high-risk systems under the European regulatory framework.

10 Research Roadmap

The Geometric Framework opens a suite of diagnostic and corrective research directions that the preceding sections have identified but not yet resolved. This roadmap consolidates these opportunities into a structured agenda organized by methodological readiness, from near-term validation experiments that can be conducted with existing infrastructure to longer-horizon challenges that require architectural generalization.

10.1 (Near-Term) Validating Security and Robustness Primitives

The structural hypotheses of Section 7 (Geometric Leakage and Topological Deviation) are the most immediately testable directions in the roadmap, as the necessary experimental infrastructure already exists in the form of adversarial attack frameworks and OOS benchmarks.

- **Empirical Validation of Geometric Leakage:** The hypothesis that the $ID - FD$ gap constitutes an adversarial attack surface requires direct stress-testing. Future work should measure Geometric Leakage across models trained with varying capacity, then correlate leakage magnitude with adversarial success rates. A confirmed correlation would provide an empirical grounding for Distillation efficiency Ω as a candidate structural robustness metric.
- **Benchmarking Topological Deviation for OOS detection:** Evaluate complementarity of trajectory-based OOS detection against confidence-thresholding. Benchmarks on the CLINC150 OOS split should report AUC for both methods individually and in combination, to determine whether geometric signatures identify precisely those anomalous inputs that confidence scores miss.

10.2 (Near-Term) Mechanistic Traceability and Targeted Oversight

The mechanistic failure decomposition of Section 6 carries operational implications requiring empirical validation.

- **Stability of Surgical Interventions:** The proposal that Readout Misalignment can be corrected by adjusting the readout vector r_j without full-model retraining must be tested in controlled experiments. Two conditions must be shown: first, that targeted adjustments do not cause collateral degradation in other intent classes whose decision boundaries share geometric proximity. Second, that the corrected model maintains structural integrity rather than introducing hidden vulnerabilities elsewhere in the manifold.

10.3 (Medium-Term) Fairness Pathology Validation

Building on the pathologies of Section 8, this track focuses on fairness diagnostic.

- **Simulation and Calibration.** Given the scarcity of demographically labeled NLU data, we propose a simulation using controlled character-level perturbations, and treating these as synthetic demographic proxies. This simulation approach would validate the detection protocol and calibrate the discrimination thresholds required to detect the presence of structural pathologies prior to deployment.
- **Geometric Regularization.** If the previous simulation study confirms the predicted signatures, work should prioritize training objectives that incorporate geometric constraints (e.g. penalizing structural asymmetries like Silhouette Score variance) directly into the model’s loss function.

10.4 (Medium-Term) Cross-Lingual Extension and Sustainable Compression

- **Typological Generalization:** To validate Resolution stability, the research must extend the typological sample to include Semitic (e.g. Arabic) and tonal language (e.g. Mandarin) to ensure the framework holds across diverse morphological formats.
- **The FD -guided compression benchmarks:** experiments must quantify the per-class performance degradation when a train model is pruned to rank FD . This will establish FD as a reliable, principled ceiling for model compression, enabling Green AI deployment on edge devices without compromising safety

10.5 (Long-Term) Extension to Transformer Architectures

The most consequential challenge for the framework’s practical relevance is the transition from RNNs to Transformer Architectures. As transformers lack explicit sequential recurrence, the trajectory interpretation established

for RNNs does not apply directly, it must be adapted. Developing an appropriate analog requires understanding whether the geometric regularities observed in RNN hidden states emerge in the multi-head attention representations of Transformer-based intent classifiers, and if so, what the correct operationalization of "trajectory" is in that context. This constitutes the central long-term challenge for extending the geometric toolbox toward relevance in contemporary deployment contexts.

11 Conclusion

This paper argues that the Geometric Framework for Intent Detection offers a necessary mechanistic lens to address the Accountability Gap inherent in high-risk NLU systems. Our central claim is not that geometric analysis should replace established behavioral audits or post-hoc XAI methods; rather, it should serve as a complementary layer of description. While current tools describe what a model attends to, the geometric perspective provides a verifiable structural record of how the model computes its decision.

The proposed diagnostic signatures are grounded in the measurable regularities of the hidden state manifold of trained RNNs, specifically the formation of stable attractor basins and their precise angular alignment with readout vectors. These structural facts allow us to move beyond binary error flags to a more nuanced mechanistic failure decomposition. We have proposed that classification failures can be traced to two distinct origins: Geometric Collapse, indicating a fundamental representational deficiency in the data, and Readout Misalignment, suggesting a model-level calibration error that might be corrected through surgical intervention rather than expensive retraining.

This paper introduces a set of structural pathologies that extend this diagnostic capability to the critical domains of robustness and fairness. By monitoring Geometric Leakage and Topological Deviation, auditors can identify the internal fragility of a model's stability before it results in a silent failure or an out-of-scope misclassification. Furthermore, our framework formalizes Mechanistic Fairness through three specific pathologies: Trajectory Dispersion, Representational Contraction and Manifold Displacement which treat algorithmic bias as a diagnosable representational instability in the hidden states of minority groups.

The diagnostic vocabulary and metrics introduced here are presented as research hypotheses, not finished auditing methodologies. Their practical utility and operational reliability must be rigorously demonstrated through the empirical milestones outlined in our Research Roadmap. This roadmap is designed to transition these primitives from controlled validations toward the complex requirements of real-world high-risk certification. By integrating this geometric layer into the broader safety evidence base, we believe the NLU community can move closer to the goal of verifiable and mechanistically grounded AI, fulfilling both the spirit and the letter of the mandates required for high-stake systems.

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